Receiver geolocation from propeller-aircraft Doppler signatures and ADS-B trajectories

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Abstract. Light aircraft radiate a narrow-band tone at the propeller blade-passage frequency. At a stationary microphone, the received frequency of that tone follows a Doppler curve fixed by the aircraft speed, the closest-approach distance and the time of closest approach. Most aircraft broadcast their own trajectories through ADS-B, so a set of such curves constrains the position of the microphone. We describe an open implementation of this inversion. It extracts tonal ridges from a whitened spectrogram, fits each flyover with a retarded-time Doppler model, enumerates closest approaches on a terrain-following grid, applies epoch-consistency and silence constraints, and refines the result by maximum likelihood, treating wind and sound speed as nuisance parameters. We test it on twelve synthetic scenarios with physically consistent audio and ADS-B data, a mis-specified atmosphere and a recording-clock offset of up to 9 min. In eleven scenarios the receiver is located within 13–87 m (median 29 m) and the clock is recovered to within 1.1 s. In the twelfth only one usable flyover is detected; the position is then under-determined, and the estimator flags it as ambiguous. We discuss how much one, two and three flyovers constrain the position, the twin solutions this produces, and the factors expected to degrade performance on recorded data.

1 Introduction

The rotating blades of a propeller produce periodic pressure fluctuations at the blade-passage frequency (BPF) $f_0 = B\Omega/60$, where B is the number of blades and Ω the shaft speed in rpm, together with its harmonics $h f_0$ [3]. General-aviation propellers give tones of roughly 50–150 Hz. A two-blade propeller at 2400 rpm radiates at 80 Hz; a four-blade turboprop propeller at 1650 rpm radiates at 110 Hz. These tones propagate over several kilometres. In a long-window spectrogram they appear as thin lines that sweep downwards as the aircraft passes, even when they cannot be heard in the recording.

Ferguson [1] showed that the time–frequency curve observed by a single sensor determines the aircraft speed, its height at closest approach and the source frequency, and Ferguson and Quinn [2] refined the estimation. The inverse problem treats the aircraft trajectory as known and solves for the observer. Trajectories of most aircraft are broadcast via ADS-B and openly archived [5].

This note describes an implementation of that inversion and evaluates it against known ground truth. It covers:

- the retarded-time Doppler model, including uniform wind, with a closed form for straight flight (Sec. 2);
- the candidate search, in which aircraft that were *not* heard also act as a constraint (Sec. 4.4);
- a synthetic evaluation in which audio and ADS-B data are generated from the same trajectories (Secs. 3 and 5);
- source code and an executable notebook that reproduce every number and figure.

All aircraft in the scenarios are fictitious.

2 Signal model

2.1 Retarded-time Doppler shift

Let τ be the time at which sound is emitted and t the time at which it reaches the receiver. In still air with sound speed c, the two are related by the slant range $R(\tau)$ at emission:

$$t = \tau + R(\tau)/c. \tag{1}$$

Phase is conserved along the ray, so the received frequency is

$$f(t) = \frac{f_0}{\mathrm{d}t/\mathrm{d}\tau} = \frac{f_0\, c}{c + v_r(\tau)}, \qquad v_r = \frac{\mathrm{d}R}{\mathrm{d}\tau}, \tag{2}$$

with the radial velocity v_r evaluated at the emission time and positive for a receding source. Evaluating the geometry at the arrival time instead (the *instantaneous* model) produces a symmetric curve. For $v = 110\,\mathrm{m\,s^{-1}}$ and $d = 0.5$ km that curve deviates from (2) by up to 11 Hz near closest approach (Fig. 1c, d).

2.2 Straight, constant-speed flyby

For a straight path flown at speed v and passing at minimum slant distance d at emission time τ_{ca}, we have $R^2 = d^2 + v^2(\tau - \tau_{\mathrm{ca}})^2$. Substituting into (1) gives a quadratic whose causal root is

$$\tau - \tau_{\mathrm{ca}} = \frac{c^2 T - \sqrt{c^2 v^2 T^2 + (c^2 - v^2) d^2}}{c^2 - v^2}, \quad T = t - \tau_{\mathrm{ca}}. \tag{3}$$

Equations (2)–(3) describe the received curve with four parameters $(f_0, v, d, \tau_{\mathrm{ca}})$. Each is visible in the curve:

$$\frac{f_+ - f_-}{f_+ + f_-} = \frac{v}{c}, \tag{4a}$$

$$\left.\frac{\mathrm{d}f}{\mathrm{d}t}\right|_{t_{\mathrm{ca}}} = -\frac{f_0 v^2}{c\, d}, \tag{4b}$$

$$t_{\mathrm{ca}} = \tau_{\mathrm{ca}} + d/c, \tag{4c}$$

Table 1. Reference scenario (random seed 7).

Duration of traffic	3.0 h
Aircraft (piston / turboprop / jet)	50 / 24 / 14
Terrain elevation	314–528 m
Recording	20 min at 4 kHz
True air temperature	6.0 °C (c = 334.9 m s^{-1})
True wind	6.0 m s^{-1} from 240°
Assumed air temperature	10.0 °C (c = 337.3 m s^{-1})
Assumed wind	none
ADS-B	223 749 reports, 88 aircraft
Clock prior / true offset	±900 s / +521 s

where $f_{\pm}$ are the approach and departure plateaux and t_{ca} is the time at which the closest-approach sound arrives (Fig. 1a, b).

2.3 Uniform wind and arbitrary trajectories

In a uniform wind $\mathbf{w}$ the wavefront is advected with the air. The travel time T for source–receiver separation $\mathbf{D}$ then satisfies $|\mathbf{D} - \mathbf{w}T| = cT$, and

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = 1 - \frac{\mathbf{k}\cdot\mathbf{v}}{c^2T + \mathbf{k}\cdot\mathbf{w}}, \qquad \mathbf{k} = \mathbf{D} - \mathbf{w}T, \tag{5}$$

which reduces to (2) when $\mathbf{w} = 0$. For turning and climbing trajectories, (1) and (5) are evaluated numerically on a 0.5-s grid of emission times. On a straight path the numerical evaluation agrees with (3) to 4×10^{-16} (relative). An 8 m s^{-1} wind changes the received frequency by up to 0.65 %.

3 Synthetic scenario

The evaluation uses synthetic scenarios so that the result can be compared with ground truth. Audio, ADS-B and truth are all derived from one set of three-dimensional trajectories (Table 1).

- *Terrain and traffic.* Sinusoidal terrain covers 150 × 150 km and is represented in a local East–North–Up frame, which includes Earth curvature. Propeller aircraft fly gently manoeuvring paths, two aircraft fly circuits at an aerodrome, and jets fly at cruise altitude.
- *Audio.* A receiver 1.5 m above ground records for 20 min at 4 kHz. Each aircraft's tone (the BPF plus three harmonics, with 0.3 % RPM wander) is rendered by evaluating the emitted phase at $\tau(t)$, using (1) and (5) with the true atmosphere. The level includes spherical spreading, atmospheric absorption and propeller directivity. The background combines pink and brown noise, gusty low-frequency wind noise, mains hum at 60/120/180 Hz and broadband jet noise.
- *ADS-B.* Reports are irregular (0.4–1.6 s apart) with occasional dropouts of 5–40 s and 6 m of GNSS position noise. Altitude, ground speed and vertical rate are quantised as broadcast: 25 ft, 0.1 kt and 64 ft min^{-1}.
- *Estimator.* The estimator assumes 10 °C and still air, and is given a recording start time uncertain by ±15 min.

Table 2 lists the propeller passes that occur during the reference recording. The estimator does not see this information.

Table 2. True propeller passes during the reference recording (withheld from the estimator). t_{ca}: arrival time of the closest-approach sound after the start of the recording.

	Type	t_{ca} (s)	d (m)	v (m s^{-1})	BPF (Hz)
P1	SR22	186.8	6070	76.9	134.4
P2	PC12	203.9	1496	112.5	103.9
P3	C172	863.3	1802	52.6	84.0
P4	SR22	910.3	5866	84.5	129.7

4 Method

4.1 Tonal ridge extraction

The recording is resampled to 1 kHz. A short-time Fourier transform is computed with a 2-s Hann window, 0.5-s hop and fourfold zero padding (0.125-Hz bins), over 30–400 Hz. Two medians are then removed: each bin's median over time in decibels, which suppresses stationary lines such as mains hum, and each frame's median over frequency, which suppresses broadband transients. What remains is the spectral contrast (Fig. 2).

Ridges are traced with hysteresis:

1. a trace starts from a seed at least 11 dB above the background;
2. it is followed forwards and backwards in a window centred on a linear prediction of the frequency, widened in proportion to the local slope;
3. within the window, the local maximum nearest the prediction is chosen in preference to the strongest, which keeps the trace on its own line where two lines cross;
4. pixels already claimed by a stronger ridge are skipped;
5. a trace ends when its running contrast falls below 6.5 dB, and fragments broken at steep sections are rejoined.

Ridges whose frequency ratio is a constant small integer are grouped as harmonics of one propeller. If a group fits poorly, the ridge whose removal most improves the fit is detached, since it most likely belongs to another aircraft.

4.2 Single-flyover estimation

Each harmonic group, with frequencies divided by harmonic number, is fitted with (2) and (3). The fit is a robust non-linear least-squares fit (soft-L_1 loss, 0.4-Hz scale) started from 15 initial guesses derived from (4). Parameter uncertainties come from the Gauss–Newton covariance, inflated by $\sqrt{4}$ because frames overlapping by 75 % are not independent.

A flyover is used only if all of the following hold:

- its ridge extends at least 8 s on either side of closest approach;
- the RMS residual is below 1.5 Hz;
- no parameter lies at a bound;
- $\sigma_d < d/2$;
- $v > 33$ m s^{-1}, the lowest plausible speed for a light aircraft in flight;
- $d < 8$ km, the search range.

In the reference recording, three of eleven detections pass (Table 3, Fig. 3). The others are rejected as partial (6) or at a parameter bound (2). The two close flyovers are recovered to about 2 % in speed and 5 % in distance. For the distant flyover

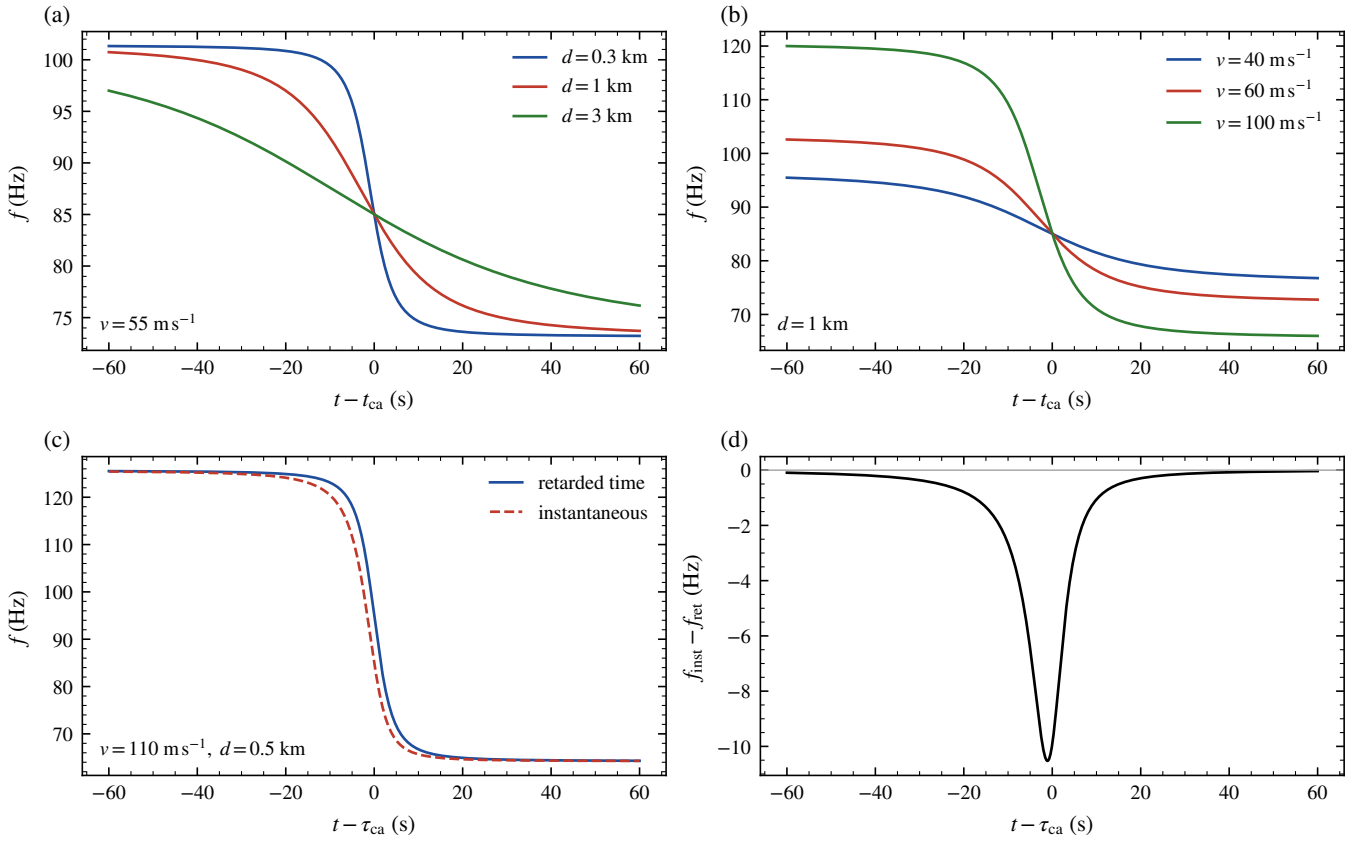


Figure 1. Received blade-passage frequency for a straight, level flyover with $f_0 = 85$ Hz. (a) The slope at closest approach scales as $1/d$ ($v = 55\,\mathrm{m\,s^{-1}}$). (b) The frequency excursion scales with v/c ($d = 1$ km). (c) Retarded-time model (2) compared with the instantaneous model for $v = 110\,\mathrm{m\,s^{-1}}$, $d = 0.5$ km. (d) Their difference.

Table 3. Usable flyovers in the reference recording: fitted parameters ($\pm 1\sigma$) and the corresponding true values (Table 2).

	$\hat{t}_{\mathrm{ca}}$ (s)	$\hat{f}_0$ (Hz)	$\hat{v}$ ($\mathrm{m\,s^{-1}}$)	$\hat{d}$ (m)	v ($\mathrm{m\,s^{-1}}$)	d (m)
F1	193.5	132.21	85.6 ± 2.6	7403 ± 536	76.9	6070
F2	204.1	104.25	114.8 ± 0.3	1570 ± 14	112.5	1496
F3	863.5	83.99	52.2 ± 0.6	1745 ± 62	52.6	1802

F1 the curve is shallow and constrains d weakly: speed and distance are overestimated by 11 % and 22 %, within 3.3σ and 2.5σ respectively.

4.3 Trajectory reconstruction

ADS-B reports are transformed into the same East–North–Up frame. Geometric altitude is ellipsoidal, so the terrain model also uses ellipsoidal heights.

- Piston and turboprop aircraft are retained by ICAO type designator; unknown types are kept as well.
- Each axis is smoothed with a Rauch–Tung–Striebel smoother [4]. A constant-velocity model fuses GNSS positions ($\sigma = 15$ m) with GNSS velocities ($\sigma = 0.7\,\mathrm{m\,s^{-1}}$).
- The result is resampled to 1 s with a cubic Hermite spline.
- Dropouts shorter than 60 s are bridged. Longer gaps split the track, and nothing is extrapolated across them.

Compared with the generating trajectories, the median reconstruction error is 1.2 m in position (95th percentile 2.6 m) and $0.18\,\mathrm{m\,s^{-1}}$ in velocity ($0.43\,\mathrm{m\,s^{-1}}$).

4.4 Candidate search

The search area (60×60 km) is divided into 200-m cells placed on the terrain surface, so slant ranges are exact in three dimensions. For every track segment within the clock prior, every local minimum of the range to every cell within 8 km is recorded as a *pass*. A parabola through the neighbouring samples refines it. Each pass stores its slant range r, speed s and arrival time t_p from (1).

Flyovers are processed in order of increasing $\hat{d}$:

- *First flyover.* A cell is admissible if it has a pass with $0.6\hat{d} \le r \le 1.6\hat{d}$ and $0.8\hat{v} \le s \le 1.25\hat{v}$ (each interval widened by 2σ), arriving within the clock prior. That pass fixes the clock epoch $t_0 = t_p - \hat{t}_{\mathrm{ca}}$.
- *Each further flyover.* It must match a different pass at the same cell with $|t_p - (t_0 + \hat{t}_{\mathrm{ca}})| \le 15$ s.

Silence constraint. Let $\mathcal{M}(\mathbf{x})$ be the passes matched at cell $\mathbf{x}$, and let $r_{\max}$ be their largest range. The cell is rejected if some other propeller pass p satisfies

$$r_p(\mathbf{x}) < r_{\max}, \qquad t_p - t_0 \in [\delta,\, T_{\mathrm{rec}} - \delta], \tag{6}$$

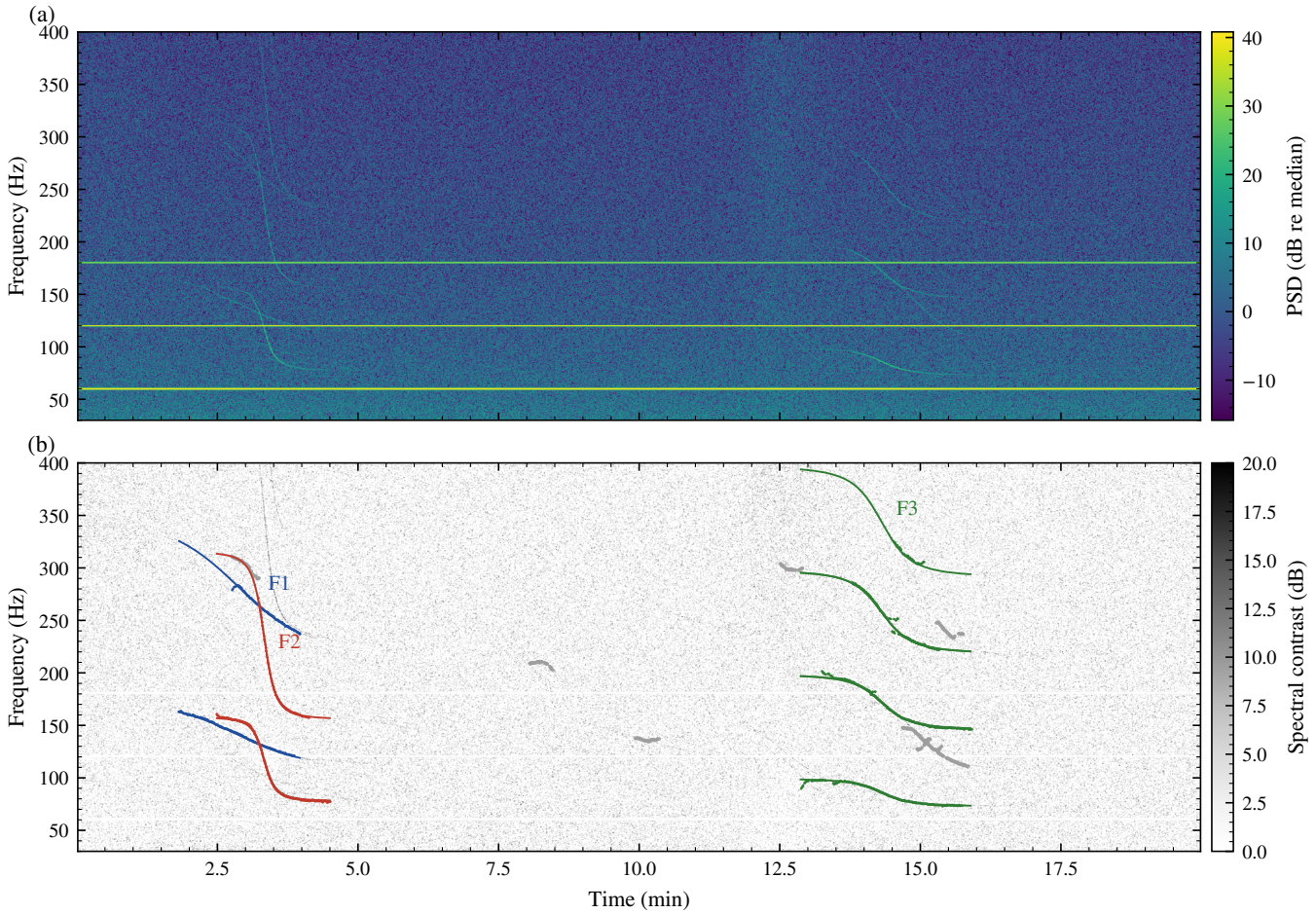


Figure 2. (a) Power spectral density of the reference recording (2-s Hann window, 0.5-s hop); the horizontal lines are mains hum. (b) Spectral contrast after median normalisation in time and frequency. Coloured markers: traced ridges of the usable flyovers F1–F3. Lines: fitted retarded-time model for each harmonic. Grey: ridges rejected as partial or implausible.

and no tonal ridge was detected near $t_p - t_0$. Here $\delta = 30$ s and T_{rec} is the recording length. Such an aircraft came closer than one that was heard, so its absence means the receiver was not at **x**.

4.5 Doppler likelihood and refinement

Each surviving hypothesis consists of a cell, an epoch and one matched trajectory per flyover. For each, the Doppler factor $g_k(t) = f/f_0$ of flyover k is predicted from the reconstructed trajectory with (1) and (5). The misfit is

$$\chi^2 = \sum_k \sum_i w_{ki} \min\left[\left(f_{ki} - \hat{f}_{0,k}\, g_k(t_0 + t_{ki})\right)^2,\ (3\,\mathrm{Hz})^2\right], \quad (7)$$

where the ridge points are (t_{ki}, f_{ki}) with contrast weights w_{ki}. The rest frequency has the closed form $\log \hat{f}_{0,k} = \sum_i w_{ki}(\log f_{ki} - \log g_k)/\sum_i w_{ki}$. The best hypotheses are refined by Nelder–Mead minimisation [6] over position and epoch, together with a uniform wind (w_E, w_N) and a sound-speed offset Δc, which carry weak Gaussian priors. Approximate confidence regions follow from $\Delta\chi^2 = 2.30$ (68 %) and 6.18 (95 %), with the residual variance taken at the optimum and divided by the window-to-hop ratio.

Table 4. Candidate reduction in the reference scenario.

	Constraint (cumulative)	Cells	Hypotheses
—	all grid cells	90 601	—
F2	range, speed, epoch admissible	17 036	18 239
+F3	common clock epoch	152	152
+F1	common clock epoch	65	65
—	silence constraint (6)	61	61
—	Doppler likelihood maximum	1	1

5 Results

5.1 Reference scenario

Table 4 and Fig. 4 show how the candidate set shrinks. In all, 1 102 529 passes are enumerated in 1.2 s. The closest flyover leaves 17 036 admissible cells. They form bands on either side of every track that passed at the right distance and at a plausible time (Fig. 5b). Requiring a common epoch with F3 leaves 152 cells, adding F1 leaves 65, and the silence constraint leaves 61. Doppler scoring then selects a single solution.

The estimate lies 15 m from the true position (Table 5, Fig. 6), and the recording clock is recovered to within 0.2 s of a prior

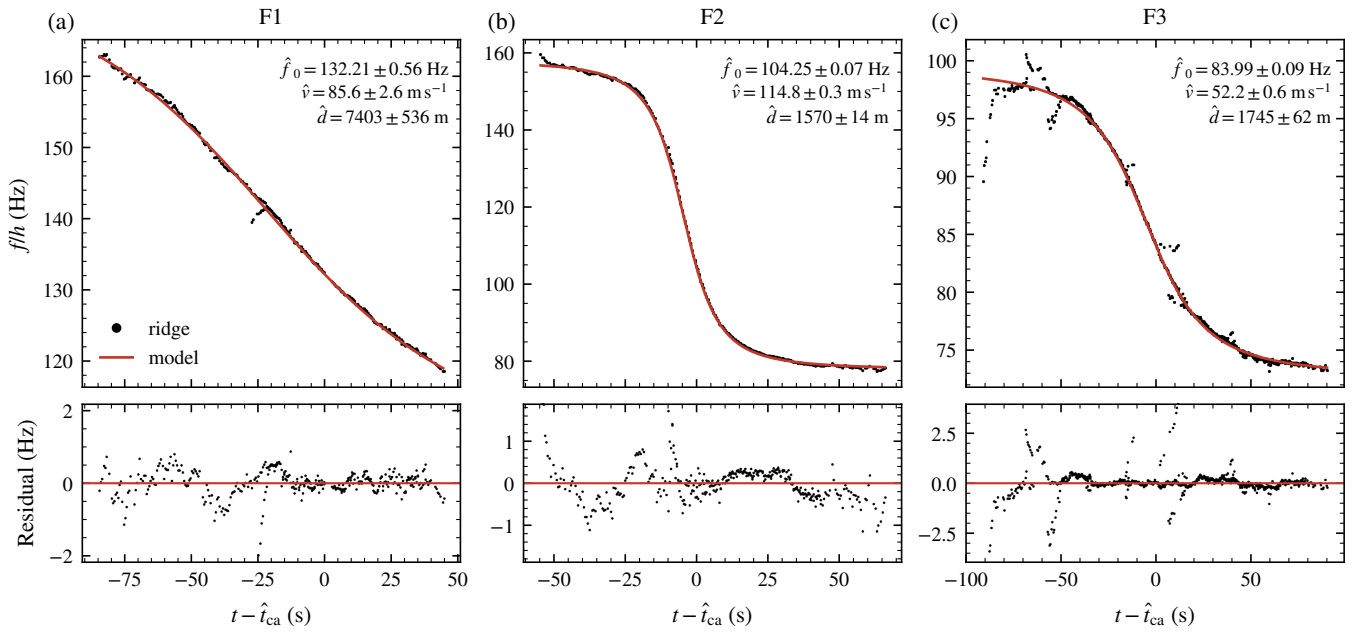


Figure 3. Single-flyover fits. Top: ridge frequencies divided by harmonic number (dots) and the fitted retarded-time model (line), with 1σ uncertainties. Bottom: residuals. The fit of F3 includes fragments at its edges where the tracer briefly followed noise; the robust loss limits their influence.

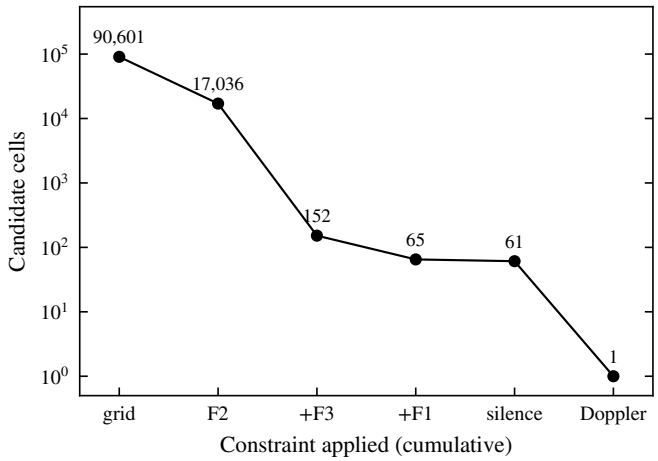


Figure 4. Number of candidate 200-m cells after each cumulative constraint (Table 4).

offset of 521 s. The best alternative more than 500 m away has a misfit 2.4 times larger (1.057 versus 0.436 Hz). Fig. 7 shows why. The alternative lies 2.9 km away. The close flyovers F2 and F3 are reproduced almost equally well at both positions. At the alternative, both aircraft pass at similar distances (1535 and 1734 m, against 1496 and 1794 m at the truth), and both closest approaches arrive about 13 s later, a shift the free epoch absorbs. For F1, the alternative implies a closest approach 1.7 km farther away and 35 s later, so only the distant flyover separates the two positions (Fig. 7a).

The fitted atmospheric parameters (3.8 °C; wind 5.2 m s^{-1} from 143°) absorb part of the model mismatch but do not recover the true values (6.0 °C; 6.0 m s^{-1} from 240°). They

Table 5. Localisation result for the reference scenario.

Estimate	38.50960° N, 98.09087° W
Truth	38.50955° N, 98.09071° W
Horizontal error	15 m
68 % / 95 % radius (statistical)	18 m / 36 m
Clock error after estimation	+0.2 s (prior offset +521 s)
RMS misfit: estimate / alternative	0.436 Hz / 1.057 Hz
Matched aircraft	F1, F2, F3 → P1, P2, P3

should be read as nuisance parameters, not as measurements.

5.2 Monte Carlo over scenarios

The full pipeline was repeated on twelve independent scenarios. Each has different traffic, terrain, receiver position and clock offset, and all share the same mis-specified atmosphere (Table 6, Fig. 8). In eleven, the receiver is located within 13–87 m (median 29 m) and the clock to within 1.1 s. In scenario 12 only one usable flyover is detected. The solution, 2.9 km from the truth, is correctly flagged as ambiguous because an alternative more than 500 m away scores within 10 % of it. The fitted temperatures range from 3.4 to 15.6 °C against a true 6.0 °C, which confirms that the atmospheric parameters are only weakly identifiable.

6 Discussion and limitations

Information content. A single flyover fixes a slant distance and an arrival time. With a known clock, the receiver lies on one of two lines parallel to the track, mirror images of each other. With an unknown clock, it can also move along those lines. A second flyover from a different direction reduces the solution set to a few points. These include *twins*, at which the

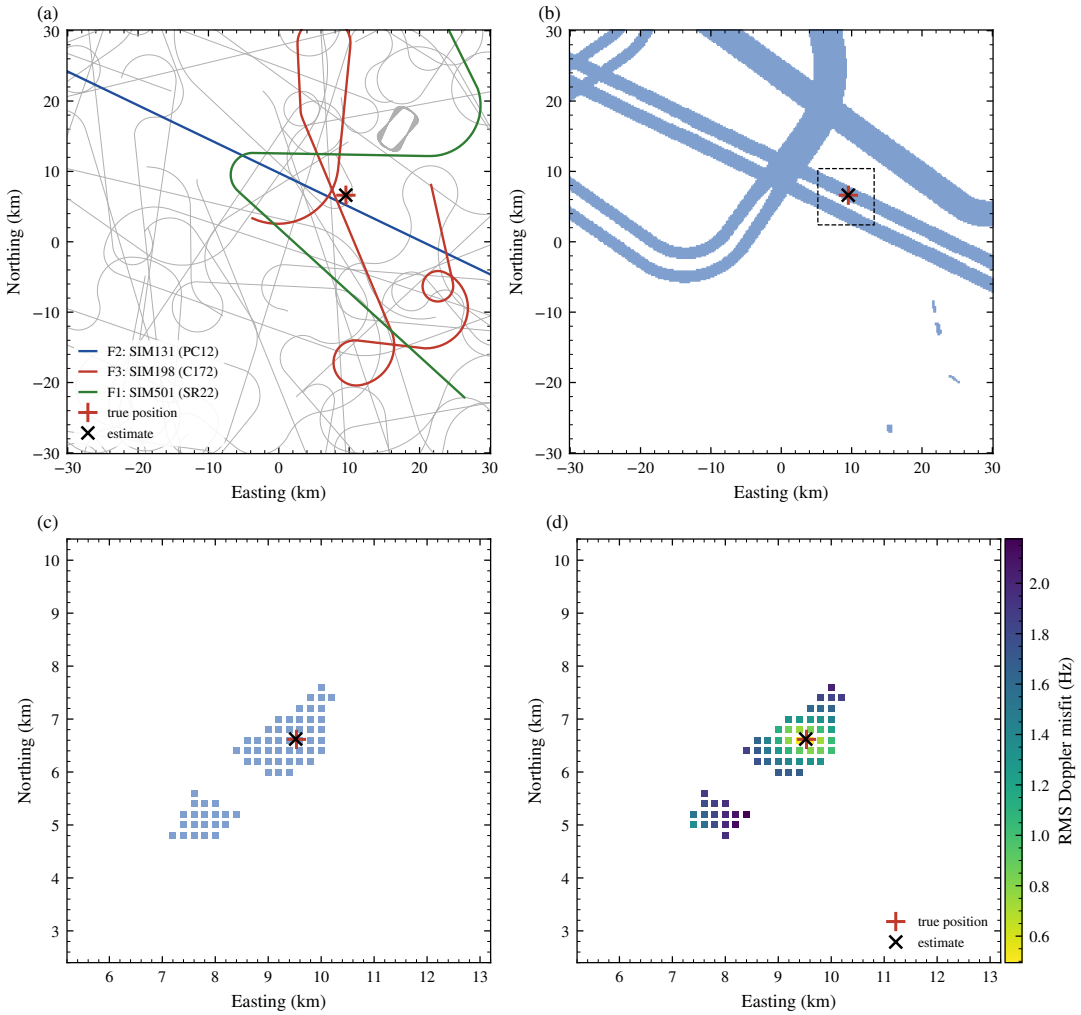


Figure 5. Spatial constraints in the reference scenario. (a) Propeller-aircraft ADS-B tracks in the time window, with the tracks matched to the observed flyovers in colour. (b) Cells admissible for the closest flyover; the dashed box marks the extent of (c) and (d). (c) Cells consistent with all flyovers under a common clock epoch. (d) Cells that survive the silence constraint, coloured by RMS Doppler misfit. +, true receiver position; ×, estimate.

Table 6. Monte Carlo over twelve scenarios. N_F: flyovers used; Δt_0: clock error after estimation; $\hat{T}$: fitted air temperature (truth 6.0 °C). A: flagged ambiguous.

Seed	N_F	Error (m)	Δt_0 (s)	χ_{rms} (Hz)	$\hat{T}$ (°C)	
1	2	63	−0.2	0.27	15.6	
2	3	87	−0.3	0.37	10.1	
3	2	19	+0.2	0.37	6.0	
4	2	25	+0.0	0.36	7.5	
5	4	71	+0.8	0.38	7.0	
6	2	83	−1.1	0.21	13.4	
7	3	15	+0.2	0.44	3.8	
8	2	27	+0.1	0.35	9.2	
9	4	53	−0.5	0.43	9.4	
10	3	29	−0.5	0.33	3.4	
11	3	13	−0.3	0.35	7.9	
12	1	2855	+0.8	0.24	9.1	A

two timing offsets are equal and are absorbed by the free epoch (Sec. 5). Twins are removed by a third flyover, by the silence constraint, or by an independent clock reference accurate to a few seconds. In the ambiguous Monte Carlo case only one flyover was usable, and the estimator reported an ambiguous result instead of a confident wrong one.

Error budget. The residual errors of 13–87 m are dominated by the mis-specified atmosphere. An error of 1 % in c scales the inferred distances by about 1 %, and wind changes the effective propagation speed by up to $|\mathbf{w}|/c$. Treating wind and Δc as nuisance parameters absorbs part of this error, but they are only weakly identifiable. The confidence regions (Fig. 6) include measurement noise only.

Detection. The hardest cases for the tracer are fast aircraft passing close by, whose frequency changes by up to 5 Hz s^{-1} at closest approach, and the crossing ridges of different aircraft. A ridge assigned to the wrong source usually fits poorly and is rejected, but the spectrogram should still be inspected.

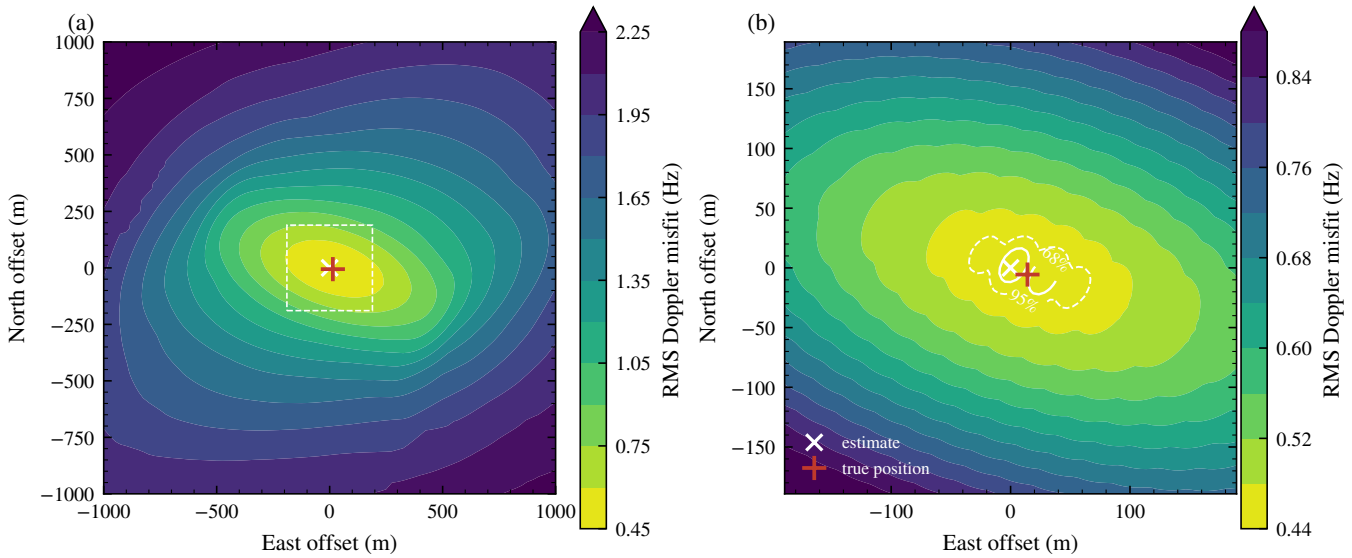


Figure 6. RMS Doppler misfit as a function of receiver position. (a) 2×2 km around the estimate; the dashed box marks the extent of (b). (b) Zoom with the approximate 68 % and 95 % confidence regions. These regions reflect measurement noise only, not model error. ×, estimate; +, true position.

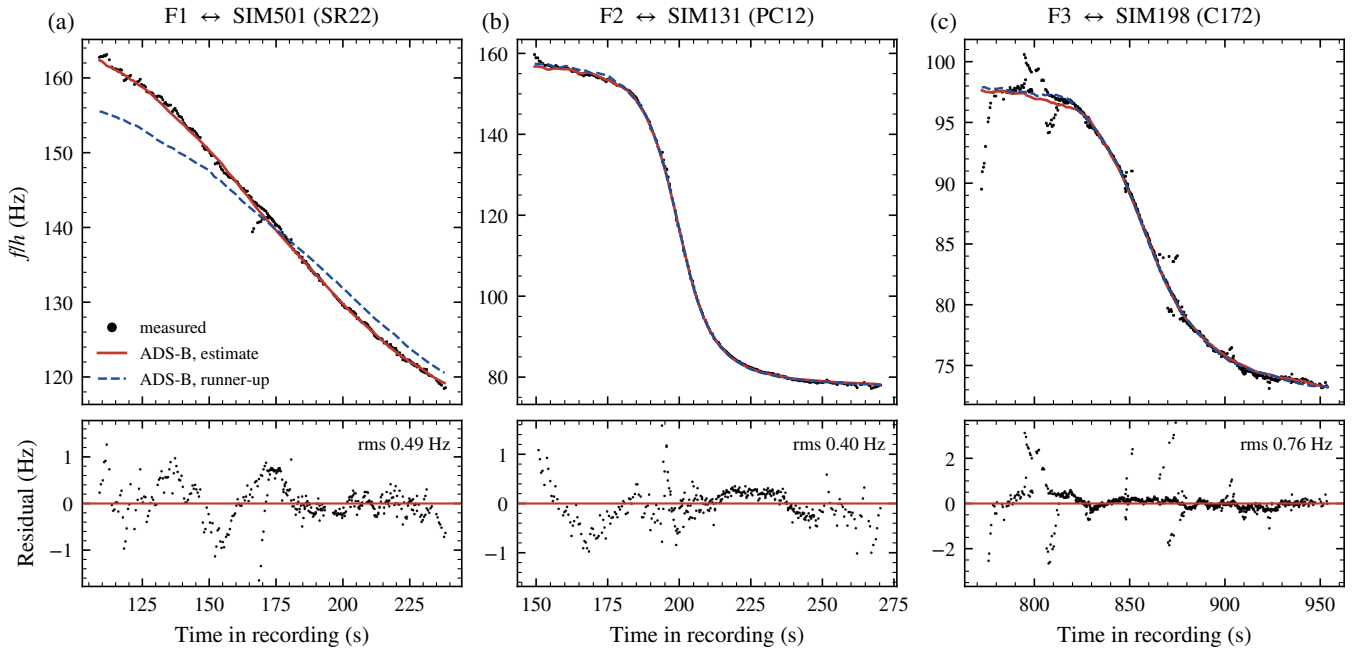


Figure 7. Measured ridge frequencies (dots) and frequencies predicted from the matched ADS-B trajectories at the estimated position (solid) and at the best alternative more than 500 m away (dashed). Bottom: residuals at the estimate.

Recorded data. Performance on real recordings is expected to be worse than on these synthetic scenarios:

- phone microphones and platform codecs often attenuate content below roughly 100 Hz and apply automatic gain control;
- the atmosphere is stratified and refracts sound, and ground reflections cause interference;
- ADS-B coverage is incomplete, and some aircraft carry no transponder.

The silence constraint is only as complete as the ADS-B data behind it.

Scope. The method requires three things: propeller aircraft passing within a few kilometres, a known date and approximate time, and ADS-B coverage of the aircraft involved. It does not apply to arbitrary recordings. The estimation of flight parameters from a single acoustic sensor has been described since at least 1992 [1, 2].

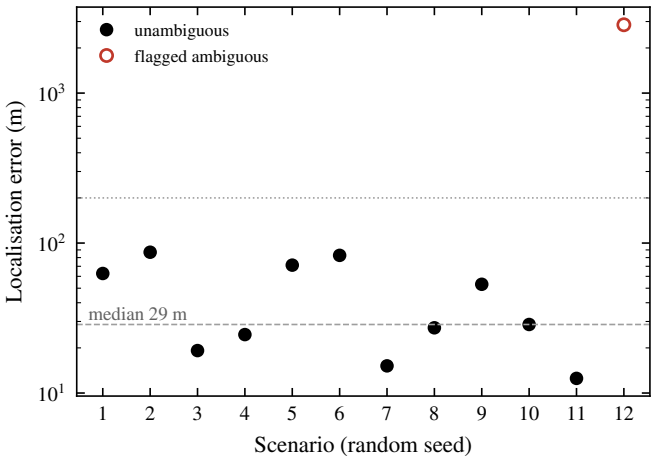


Figure 8. Horizontal localisation error for twelve synthetic scenarios. Open marker: flagged ambiguous. Dashed line: median of the unambiguous cases; dotted line: 200 m.

7 Conclusion

Doppler-shifted propeller tones combined with ADS-B trajectories can locate a receiver to within tens of metres when two or more flyovers are recorded and the date is known. In synthetic tests with a mis-specified atmosphere and an unknown clock offset of up to 9 min, eleven of twelve scenarios were located within 13–87 m. The remaining scenario, which contained only one usable flyover, was correctly reported as ambiguous. Evaluation on recorded data with independently surveyed ground truth remains to be done.

Code and data availability

The implementation (`doppler_geo` v0.2.0), the synthetic-scenario generator and an executable notebook that reproduces every number and figure in this note are provided with this manuscript. All scenario data are synthetic, and all aircraft identifiers are fictitious.

References

[1] B. G. Ferguson, J. Acoust. Soc. Am. **92** (1992), doi:10.1121/1.403934.

[2] B. G. Ferguson and B. G. Quinn, J. Acoust. Soc. Am. **96** (1994), doi:10.1121/1.410320.

[3] H. H. Hubbard (ed.), *Aeroacoustics of Flight Vehicles: Theory and Practice*, NASA Reference Publication 1258 (1991).

[4] H. E. Rauch, F. Tung and C. T. Striebel, AIAA J. **3**, 1445 (1965).

[5] adsb.lol, open ADS-B data archive (ODbL), `https://adsb.lol`.

[6] J. A. Nelder and R. Mead, Comput. J. **7**, 308 (1965).